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Implications of Next Generation Fingerprint Technologies

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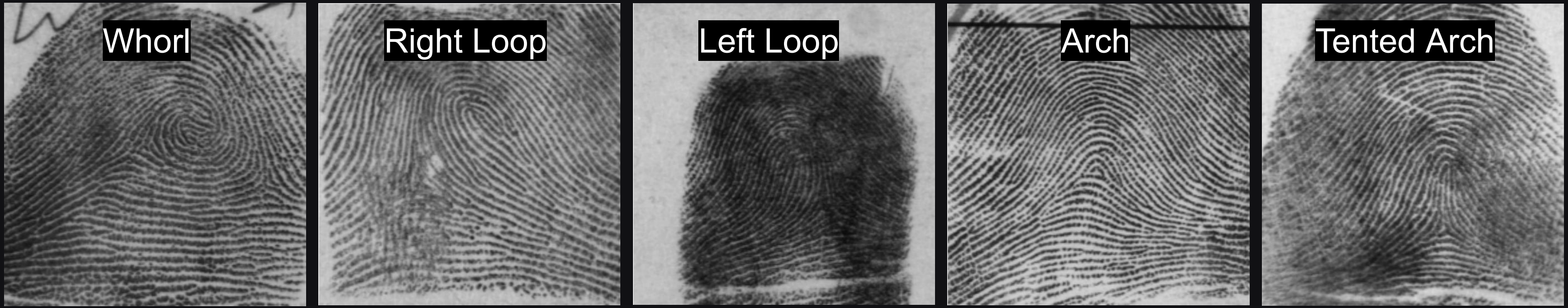
Pioneers of Fingerprints

- Evidence of fingerprints in early clay tablets and seals from 1955 BC
- Nehemiah Grew - ridges, furrows, and pore structure studied in 1624
- Sir William Herschel - fingerprints unchanged from 1860-1890 (**permanence**)
- Henry Faulds - fingerprints grow back in same pattern (**permanence**)
- Sir Edward Henry - Henry classification system of fingerprints in 1900
- Francis Galton - Identified fingerprint minutiae in 1892.

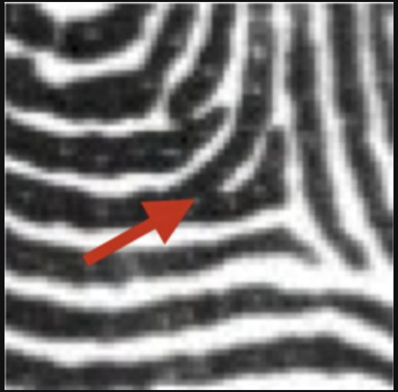


Henry Classification System

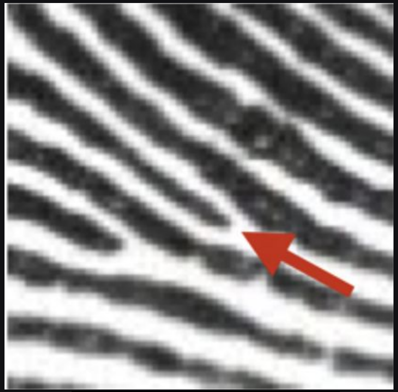
- Jain, Anil K., et al. "Fingerprint recognition." *Introduction to Biometrics*. Cham: Springer International Publishing, 2024. 75-117.
- Ashim K Datta. Advances in Fingerprint Technology. CRC press, 2001.

Fingerprint Uniqueness

- “They have the unique merit of retaining all their peculiarities unchanged throughout life, and afford in consequence an incomparably surer criterion of identity than any other bodily feature.” - Sir Francis Galton (1892) [2]
- Galton went on to describe the well known minutiae points.
- Statistical model of minutiae distributions studied in [1].



Minutiae Bifurcation



Minutiae Ending

[1] S. Pankanti, S. Prabhakar, and A. K. Jain, "On the Individuality of Fingerprints", IEEE Trans. Pattern Analysis and Machine Intelligence, 2002
[2] Galton, Francis. *Finger prints*. No. 57490-57492. Cosimo Classics, 1892.

Fingerprints Permanence

- A given fingerprint pattern maintains high similarity over time [1]
- A fingerprint pattern here is defined primarily by its extracted minutiae set.

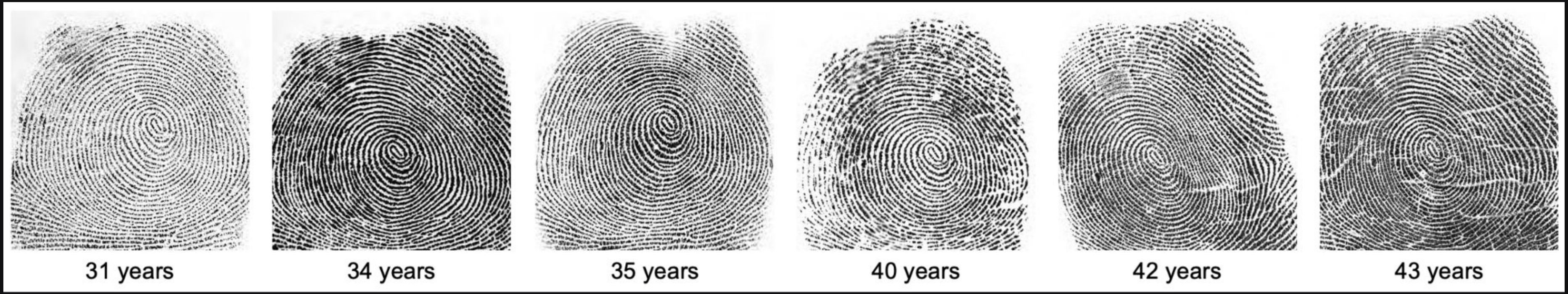
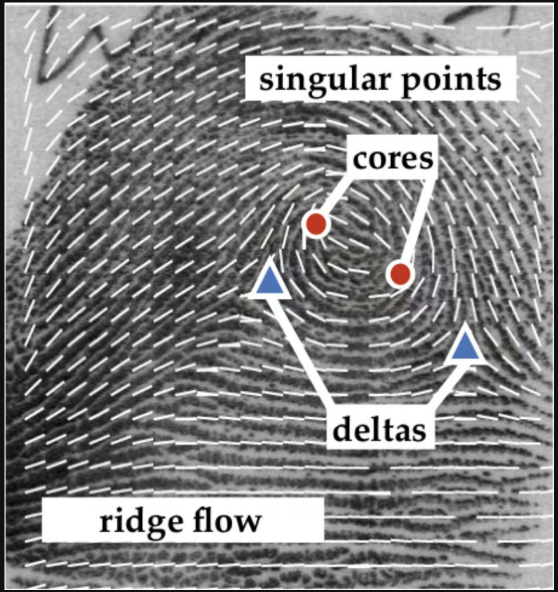


Figure taken from: Grosz, Steven A. *On the Generalization of Fingerprint Embeddings*. PhD Thesis. Michigan State University, 2024.

[1] S. Yoon and A. K. Jain, "Longitudinal Study of Fingerprint Recognition", Proc. National Academy of Sciences (PNAS), July 2015.

Fingerprint Representation

- Prevailing fingerprint representation is comprised of 3 levels of features
  - Level-1: Ridge orientation field, singularities (cores, deltas)
  - Level-2: Minutiae Set (x, y, theta)
  - Level-3: Pores, incipient ridges, islands etc.



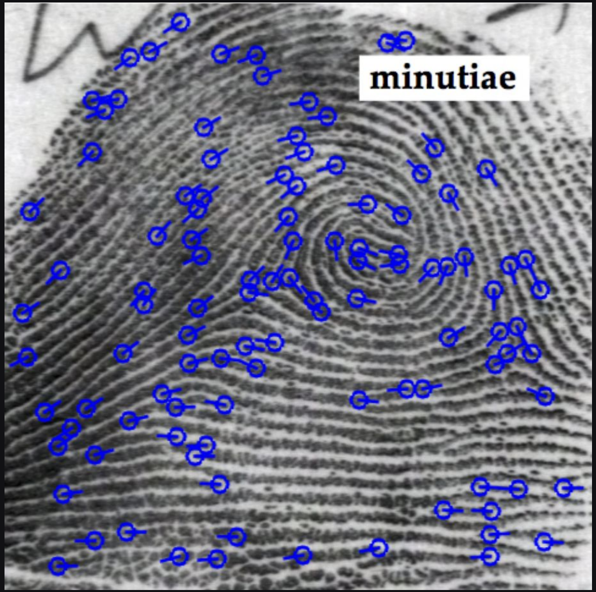
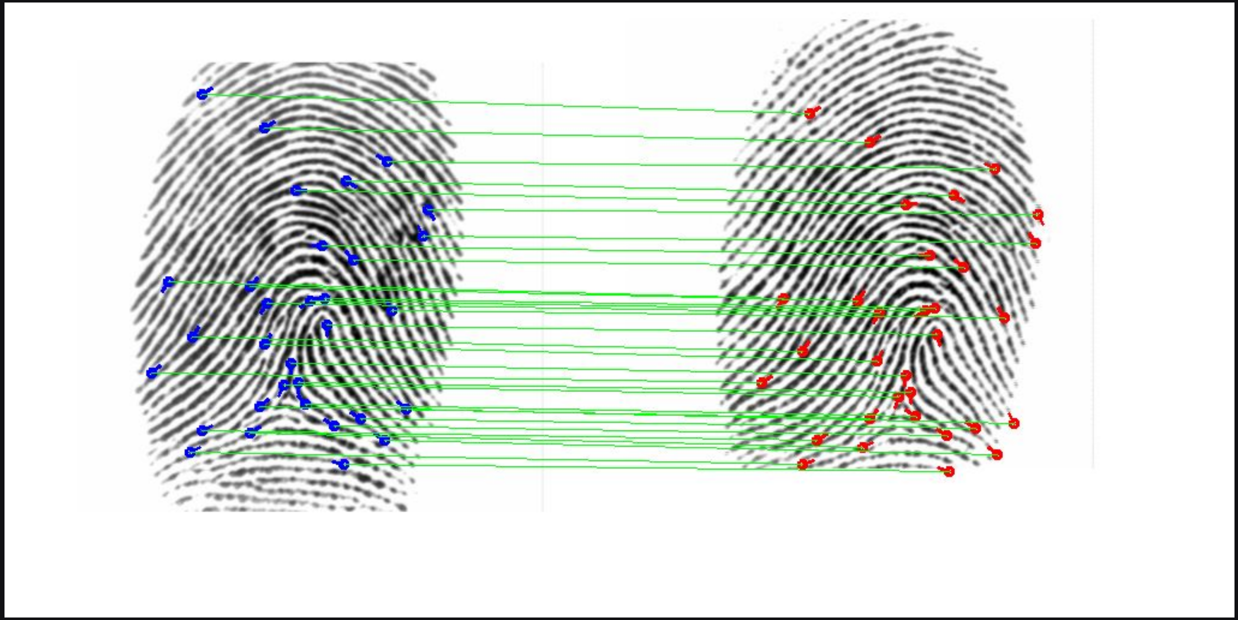


Figure taken from: Engelsma, Joshua James. *Contributions to Fingerprint Recognition.* PhD Thesis. Michigan State University, 2021.

Fingerprint Matching

- Minutiae matching tries to find corresponding minutiae points across a pair of images (minutiae sets). More correspondences leads to higher similarity scores

Example of a Successful Minutiae Match; 31 corr; similarity of 1.0 (range 0, 1)

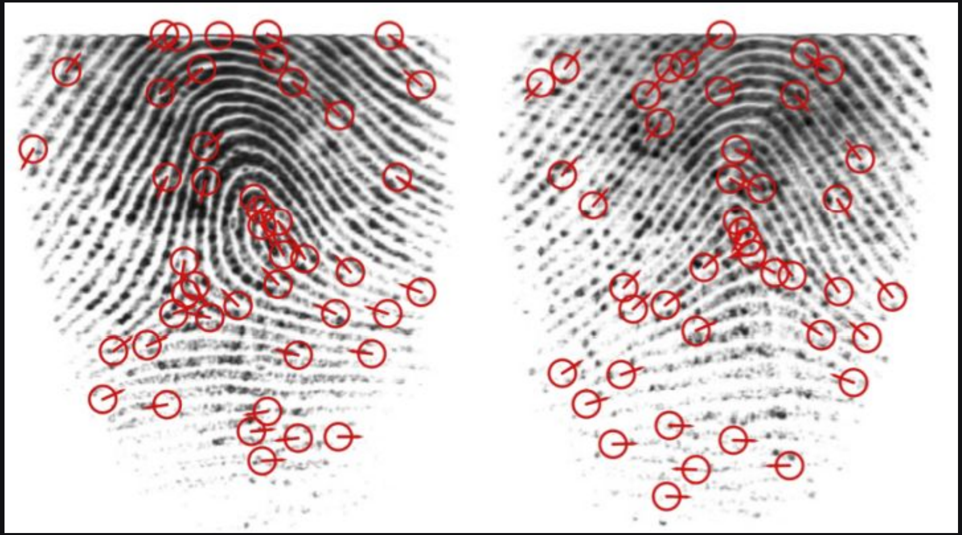


What are the Limitations of Minutiae Matching?

- Difficult cases still result in false rejects and false accepts.
  - Large non-linear distortions.
  - Occluded or missing (partial) friction ridge (wet or dry fingers).
  - Highly similar minutiae distributions.

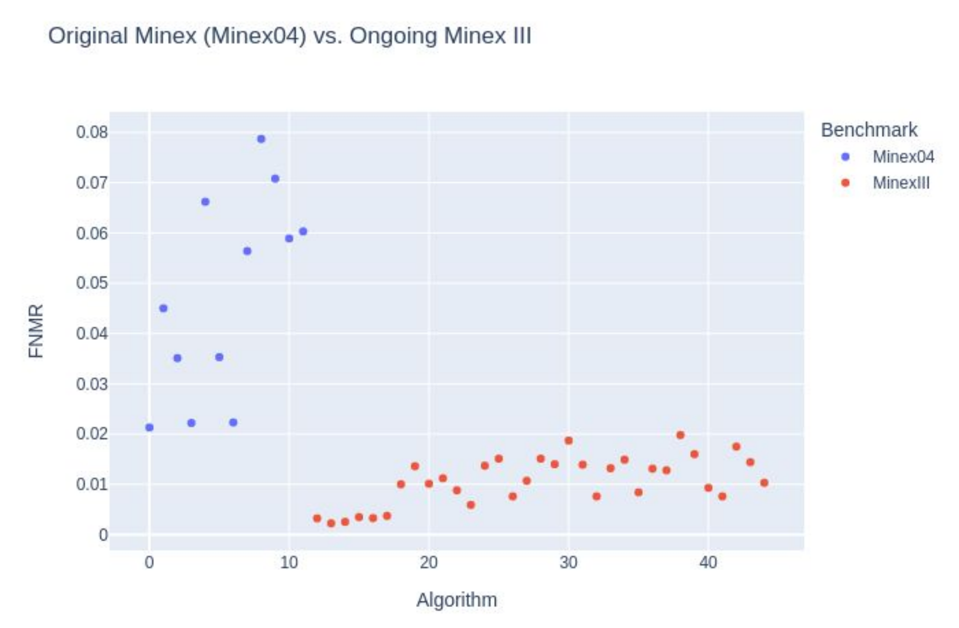
False Reject

False Accept



Minutiae Based AFIS (MINEX Performance)

- Several benchmarks continue to evaluate the performance of fingerprint recognition systems based solely on minutiae points, e.g., NIST MINEX



- The lowest error rates from Minex04 (2004) are still higher than the Minex III (ongoing) worst performing minutiae matchers.
- NOTE: Minex04 FNMR computed at FMR=1e-3, Minex III computed FNMR at FMR=1e-4; *both on Port of Entry datasets*

Results compiled from: https://www.nist.gov/itl/iad/btg/minutiae-interoperability-exchange-minex-iii and https://www.nist.gov/system/files/documents/itl/iad/ig/minex_native.pdf

Minutiae Based AFIS (MINEX Performance)

- What is the best that SOTA minutiae matching can do?
  - Below is the average error rate computed from the top-10 vendors in Minex III (ongoing). Also shown is the best overall error rate.
  - These low error rates are very impressive, but does this mean that this is the best *fingerprint* recognition is capable of?
  - Are these minutiae matching algorithms able to capitalize on the latest advancements in CV/ML over the past decade?

Average FNMR@FMR=1e-4	Best FNMR@FMR=1e-4
0.0062	0.0024

Results compiled from: https://www.nist.gov/itl/iad/btg/minutiae-interoperability-exchange-minex-iii

Face Recognition Improvements over Time

- Tremendous advances in Computer Vision and Machine Learning circa 2012 … face recognition community latches on.
  - Handcrafted and engineered features for image classification are replaced by highly parameterized Deep Networks optimized over massive training datasets.

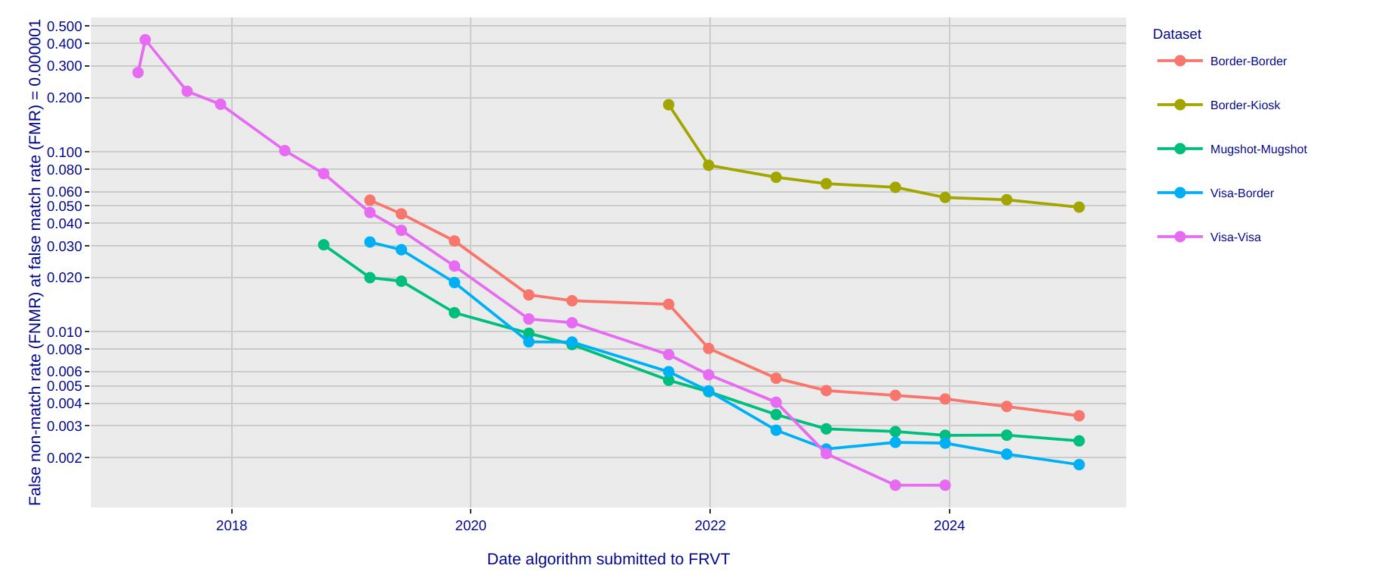
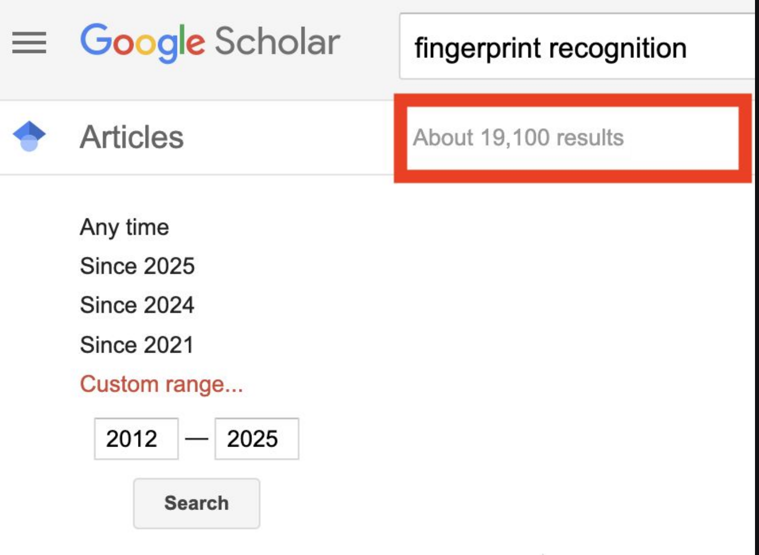
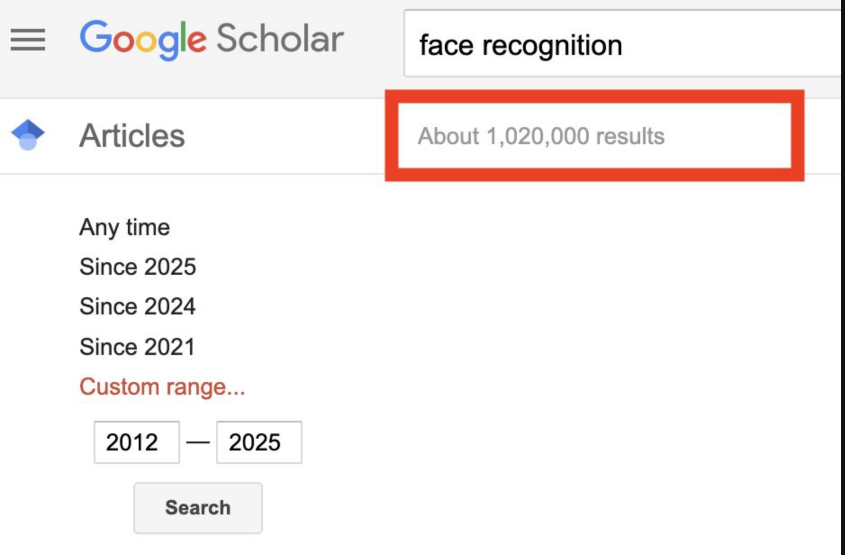


Figure retrieved from :https://pages.nist.gov/frvt/html/frvt11.html

Are Fingerprint Recognition Systems Adapting?

- Perception in academia that fingerprint recognition is a solved problem.
  - Quote from a CVPR 2019 reviewer reviewing my fingerprint paper: “Fingerprint is a solved problem! Why are you working on this!”
  - Plethora of publicly available face data for training; not so with fingerprints.

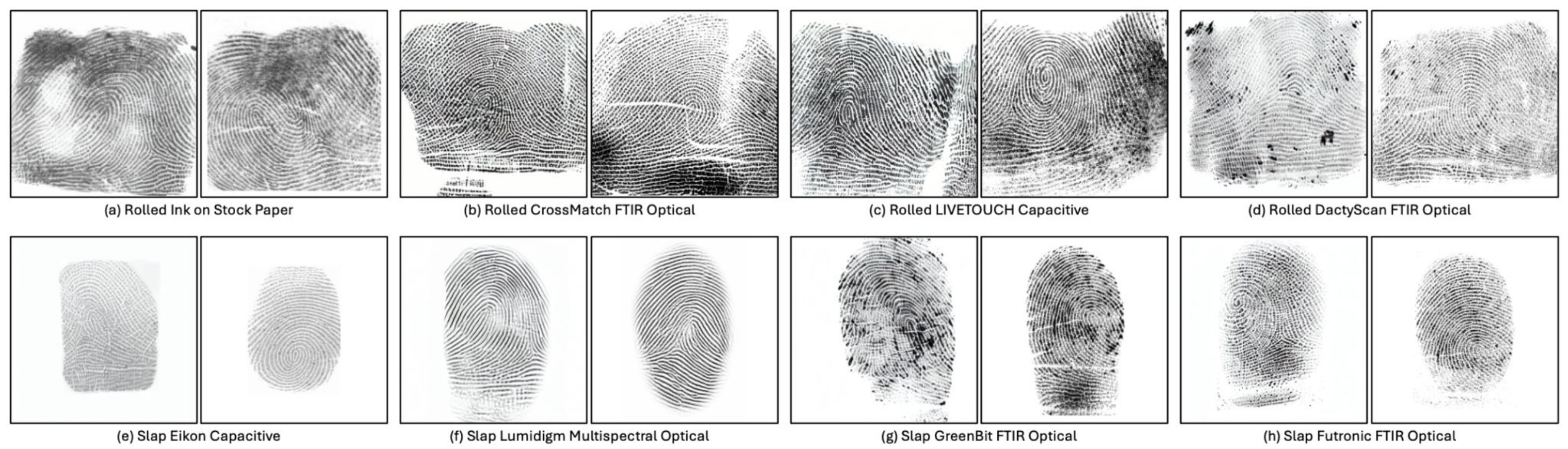




Statistics retrieved from https://scholar.google.com, April 1, 2025

Are Fingerprint Recognition Systems Adapting?

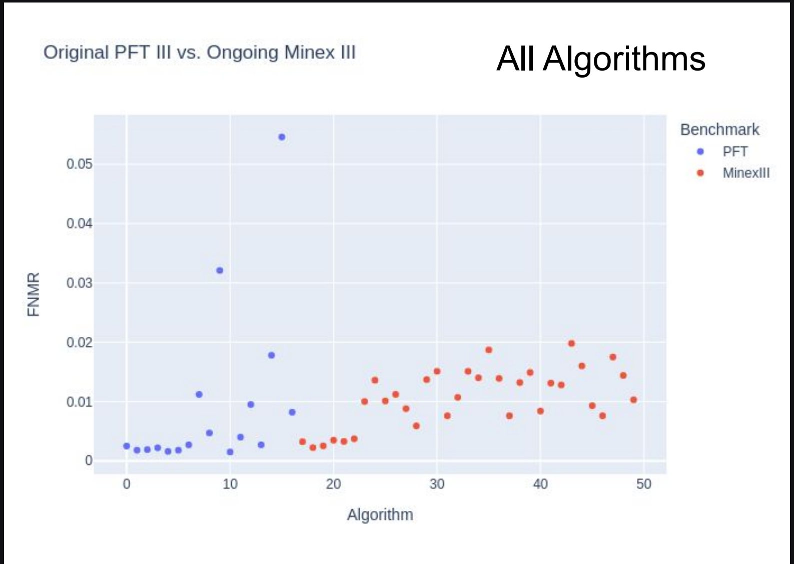
- As mentioned, paucity of publicly available fingerprint data is very limiting with respect to applying next generation technologies.
- Synthetic data may provide some answers here.

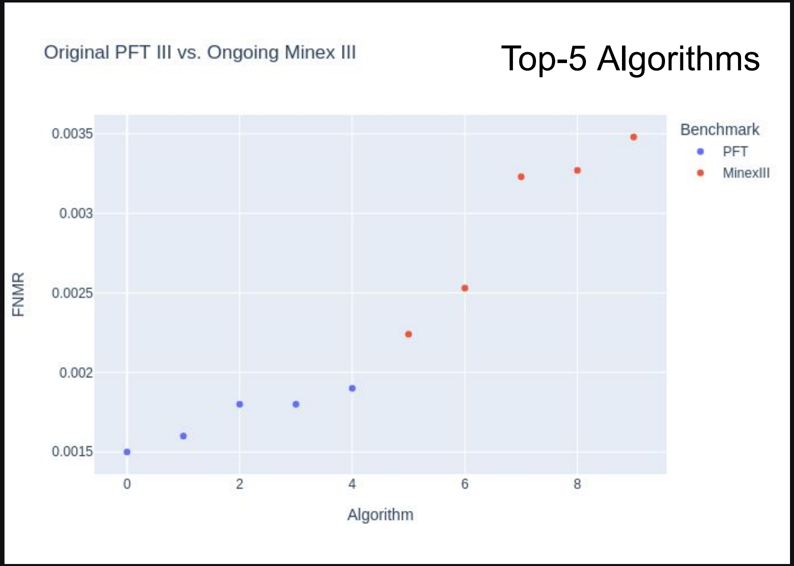


- Engelsma, Joshua James, Steven Grosz, and Anil K. Jain. "PrintsGAN: Synthetic fingerprint generator." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45.5 (2022): 6111-6124.
- Grosz, Steven A., and Anil K. Jain. "Universal fingerprint generation: Controllable diffusion model with multimodal conditions." *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2024).
- Shoshan, Alon, et al. "FPGAN-control: A controllable fingerprint generator for training with synthetic data." *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 2024.

Comparing Minutiae Templates to Proprietary Templates

- The NIST PFT (Proprietary Fingerprint Template) benchmarks fingerprint matching based on whatever features a vendor desires to extract.
  - Incredibly important benchmark! Enables better utilization of latest CV/ML advances in fingerprint recognition systems.
  - Plots compare PFT with MINEX algorithms on the same dataset (FMR=1e-4).





Results compiled from: https://www.nist.gov/itl/iad/btg/minutiae-interoperability-exchange-minex-iii and https://pages.nist.gov/pft/results/pftiii/

Comparing Minutiae Templates to Proprietary Templates

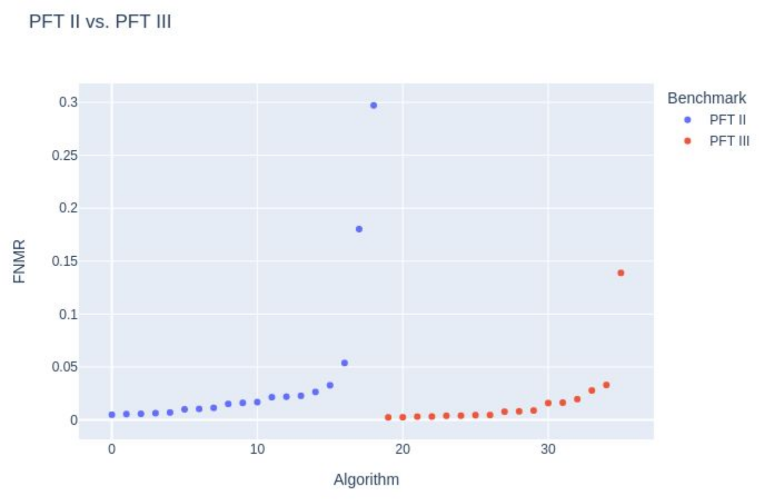
- The best NIST PFT Algorithms obtain substantially lower error rates than the best NIST Minex Algorithms on the same data.
  - The average below is computed from the top-10 vendors that compete in PFT and MINEX.
  - Using proprietary features results in > 60% decrease in error rate over minutiae only features.

Average (Best) FNMR: **Minex**	Average (Best) FNMR: (**PFT**)
0.0062 (0.0024)	0.0023 (0.0015)

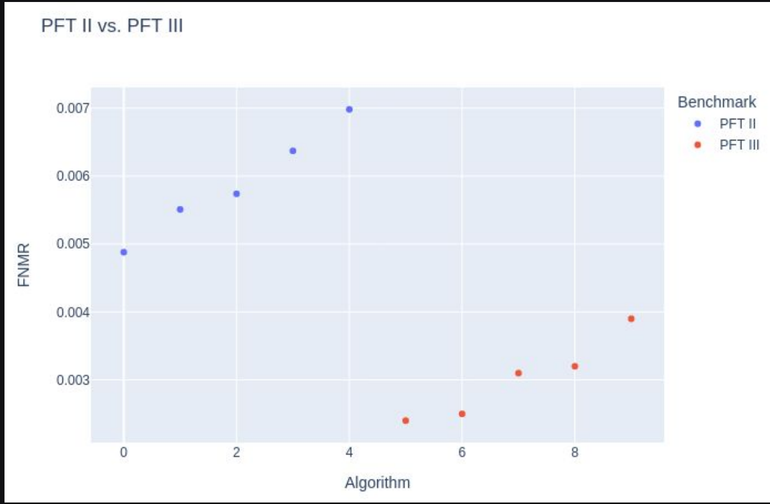
Results compiled from: https://www.nist.gov/itl/iad/btg/minutiae-interoperability-exchange-minex-iii and https://pages.nist.gov/pft/results/pftiii/

Proprietary Templates Over Time

- Proprietary templates themselves have drastically improved from PFT II (2010-2019) to PFT III (2019 - present)
  - Plots below compare PFT II vs. PFT III algorithm submissions on the AZ/LA law enforcement datasets. (FMR of 1e-4)



All Algorithms

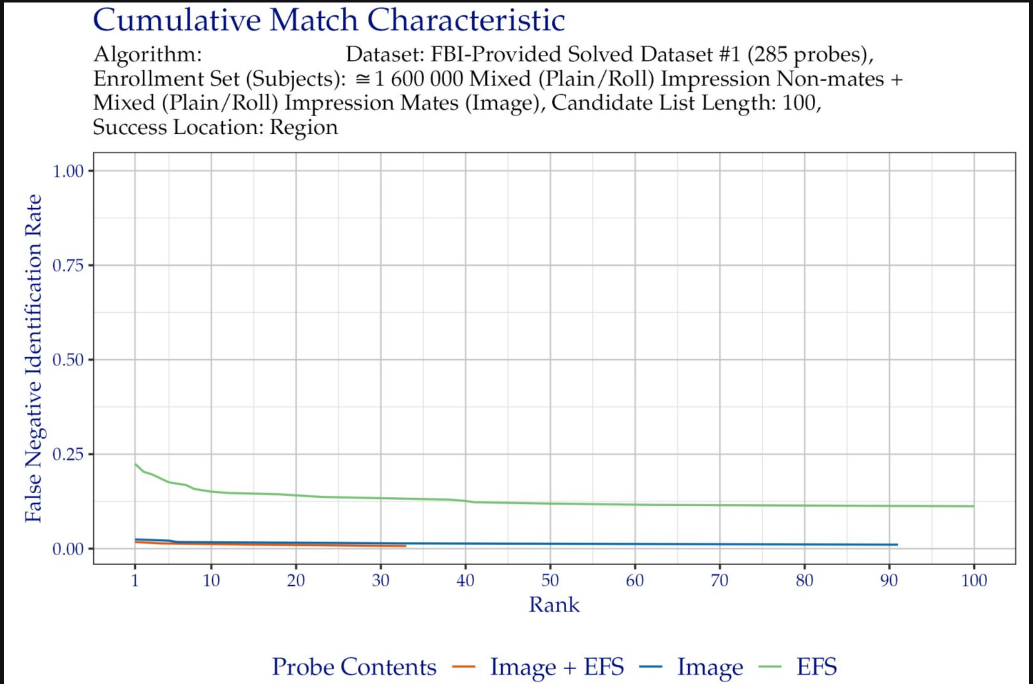


Top-5 Algorithms

Results compiled from: https://www.nist.gov/itl/iad/image-group/proprietary-fingerprint-template-evaluation-ii-pftii-results and https://pages.nist.gov/pft/results/pftiii/

Comparing Minutiae to Proprietary Templates: Latent

- The NIST ELFT (Evaluation of Latent Fingerprint Technologies) allows vendors to extract any features desired for latent identification.
  - NIST compares these proprietary latent templates with latent matching that utilizes EFS (extended feature sets: minutiae, cores, deltas etc.)



Results compiled from:
https://pages.nist.gov/elft/elft_1_x/results/

Latent Image Search over Time

- NIST has run a number of latent benchmarks. Even comparing the current benchmark with the most recent benchmark prior, we see impressive improvements.
  - The best-3 algorithms from the 2012 NIST Latent benchmark had an average rank-1 accuracy of 65%
  - The best-3 algorithms from the ongoing NIST ELFT benchmark have an average rank-1 accuracy of 88%!

Average Rank-1 Accuracy: **NIST Latent 2012**	Average Rank-1 Accuracy: **Ongoing NIST ELFT**
65%	88%

Averages from the top-3 Algorithms per Benchmark

Results compiled from: https://pages.nist.gov/elft/elft_1_x/results/ and https://www.nist.gov/system/files/documents/2016/12/19/elft-efs2_iai_2012_final.pdf

Fingerprint Comparison Speeds

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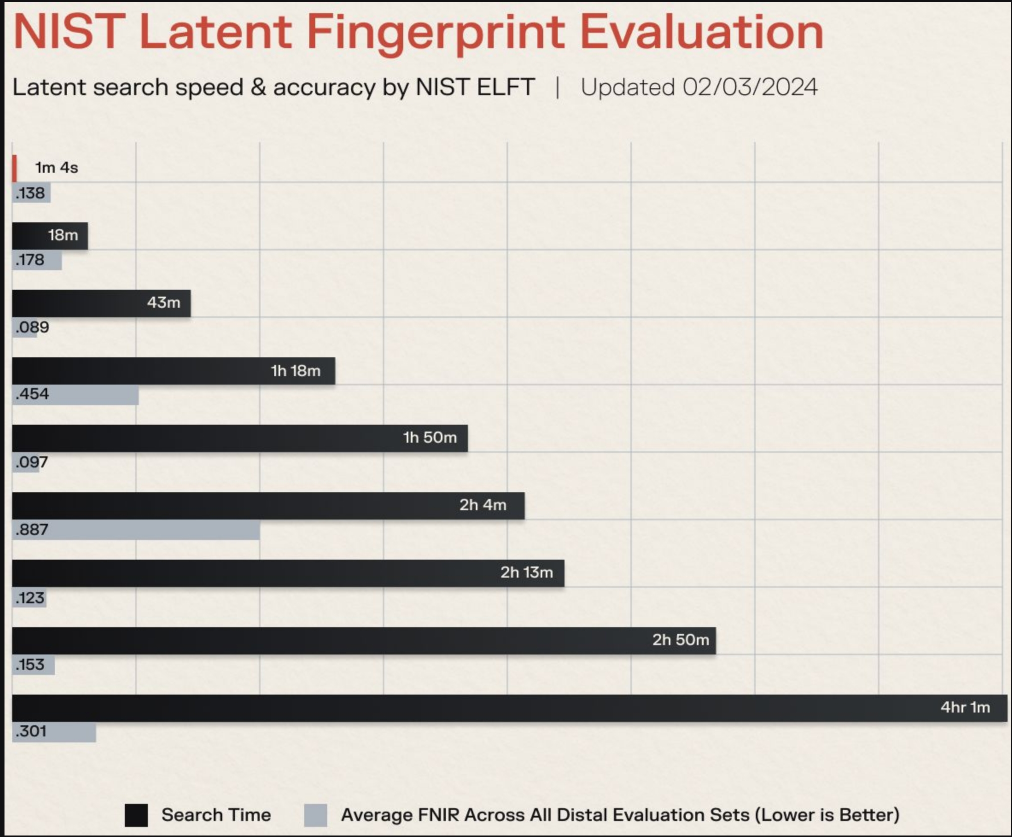
- Conventional minutiae matching algorithms are very computationally expensive!
  - Proprietary Fingerprint Algorithms are making significant strides in comparison speeds. Face is still dominating on this metric.
  - The below table shows the comparison speeds of the four most accurate algorithms in each benchmark. The bottom row shows the fastest algorithm.

Algorithm Comparison Speeds in Milliseconds

NIST MINEX	NIST PFT	NIST FRTE (Face)
5.77	4.76	0.0008
9.84	5.22	0.0001
4.14	0.29	0.06
3.57	0.70	0.01
0.37	**0.0006**	**0.00007**

Results compiled from: https://www.nist.gov/itl/iad/btg/minutiae-interoperability-exchange-minex-iii, https://pages.nist.gov/pft/results/pftiii/ , and https://pages.nist.gov/frvt/html/frvt11.html

Fingerprint Comparison Speeds: Latent Search



- Conventional minutiae matching algorithms are computationally expensive.
- The fastest NextGen latent algorithm performs search against 1.6M subjects (~30M fingerprints) gallery in 64 seconds. Other algorithms take hours.
- Real time latent search (with high accuracy) is now a reality.

Results compiled from:
https://pages.nist.gov/elft/elft_1_x/results/

NextGen AFIS

Summary of NextGen Algorithm Improvements

- Fingerprint matching based on minutiae points has a rich history. However, it's somewhat limited in its ability to leverage the latest advancements in CV/ML like face recognition has.
- Proprietary template fingerprint matching does open the possibility for companies to explore these new methods. The results speak for themselves with > 60% error reductions.
- Proprietary template latent search has improved from 65% accuracy to 88%.
- Latent search speeds have gone from hours to a single minute on a gallery of 30M.

NextGen AFIS

Implications of NextGen Fingerprint Recognition on Industry

- Which companies will innovate?
    - Companies that innovate and leverage latest CV/ML will be the future of fingerprint recognition technologies.
    - New companies will likely emerge and challenge reputable, long standing players.
    - Competition is driving and will continue to drive fast innovation in this space.
    - Rapid advances in the technology will drive the need for evergreen licensing.

NextGen AFIS

Implications of NextGen Accuracy

- What is the upper bound of fingerprint recognition performance?
    - How do state-of-the-art algorithms perform at extremely low FAR (e.g., 1e-6)? Important for large scale applications.
    - How do fingerprint recognition systems perform at scale? (FRIF E1N)
    - What failure cases are left to address? How much of a fingerprint is needed to perform a match? How many minutiae points? Still 15 minutiae?
    - Brief speculative comment: Current PFT performance likely capped by NFSeg failures.

NextGen AFIS

Implications of NextGen Accuracy

- Will minutiae point matching stand the test of time? (I strongly suspect yes, but augmented with additional features)
  - What are the impacts to forensic examiners? How can algorithms better leverage orthogonality of domain inspired features?
  - What is the impact to evidentiary value in court of law?
  - Do we need to re-think models on the “individuality of fingerprints”?

NextGen AFIS

Implications of NextGen Speed

- Can fingerprint recognition systems rival the speeds of face recognition algorithms?
    - Faster speeds means much lower cost.
    - Real time latent search opens up new investigative opportunities. E.g., searching unsolved latent file at a traffic stop.
    - What other applications can be considered with real time search speeds?

NextGen AFIS

Whats Next?

- Fingerprint recognition algorithms will continue to make rapid accuracy improvements.
- Latent Fingerprint recognition accuracy is going to keep getting better and better.
  - Levels of accuracy unthinkable even 10 years ago.
  - Cold-case hits, more new hits
  - Increased Examiner Value
- Algorithm comparison speeds will continue to get faster and faster (already algorithms in real time (e.g., 60 second search speeds).
  - AFIS cost will go down
  - Unlikely that latent fingerprint search speeds will catch face search speeds… too much complexity and nuance in the latent search pipeline.